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The role of accelerator programmes on the capital structure of start-ups

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Abstract Several factors affect the capital structure of start-ups. In this study, we compare the capital structure choices of accelerated and non-accelerated start-ups to evaluate the role of accelerated programmes on the capital structure decisions for start-ups. Using a comprehensive database provided by Emory University, we find that accelerated start-ups have higher external equity ratios than non-accelerated start-ups, after accounting for firm-specific differences and unobserved start-ups factors, particularly when the economic conditions deteriorate. We also confirm that accelerated start-ups raise more funding through philanthropic investors, but this effect disappears when we control for firm fixed effects. These results suggest that accelerators provide informative signals about the quality of the venture, albeit only to external equity investors, and particularly during recession periods. Our findings have implications for a better understanding of the role of asymmetric information and business cycles in capital structure decisions for accelerated start-ups.

Plain English Summary Accelerator programmes aid venture finance. But which sources are most likely to invest in accelerated start-ups, and how do business cycle conditions affect this relationship? Using a comprehensive database provided by Emory University, we identify the financing sources of start-ups (debt, external equity, and philanthropic capital) for the years prior to, during, and after successfully and unsuccessfully applying for an acceleration programme. We find that accelerated start-ups have higher external equity ratios than non-accelerated start-ups, particularly when economic conditions deteriorate. These results suggest that accelerator programmes provide informative signals about the quality of the venture, albeit only to external equity investors, and particularly during recession periods. The principal implication of this study is that it could be advisable for entrepreneurs who are planning to raise equity from external investors to first apply for accelerated programmes in order to improve their chance of obtaining external equity, especially during economic downturns.

Keywords Capital structure · Accelerated programmes · Start-ups · Venture finance · Debt · External equity · Philanthropic capital

JEL classification G32 · L26 · M13 · M21

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1 Introduction

The importance of start-up's initial financial capital is well established in the previous literature. Financial

capital determines the size (Bruderl et al., 1992), growth (Colombo & Grilli, 2005; Cooper et al., 1994), and survival of a start-up.[1] Ventures with larger financial resources are able to overcome temporary hardships and/or managerial mistakes more easily, acquire better resources and technology, and start operating on a larger scale. The choice of funding (e.g. own equity, external equity, or debt) can also signal the value of the start-up to investors and other external entities, and thus reduce information asymmetries and agency problems (Ross, 1977). Not only does the composition of the capital structure signal the real value of a start-up, but it also demonstrates the perspectives for growth and the level of commitment of entrepreneurs in the start-up (Brealey et al., 1977; Cassar, 2004; Greenwald et al., 1984; Myers, 2001). However, several questions remain to be answered, namely: does the capital structure differ between accelerated and non-accelerated start-ups? Can accelerator programmes mitigate the asymmetry of information between start-ups and outside investors? Do economic conditions affect the choice of capital structure for accelerated start-ups in relation to non-accelerated start-ups?

Accelerator programmes have become a common feature across many entrepreneurial ecosystems (Hochberg, 2016). These programmes are of limited duration and are designed to support start-ups in their early stages of life by providing seed funding, peer-based mentoring and coaching, and entrepreneurship education (Cohen, 2013; Pauwels et al., 2016). Accelerator programmes are usually cited as being a key contributor for the performance of a start-up (Chan et al., 2020; Hallen et al., 2020). However, whether and how accelerator programmes actually benefit entrepreneurs is still open to question.[2] For even if accelerator programmes are associated with better venture outcomes, there is no clear indication that the associated outcomes are a direct result of accelerator participation (Sørensen, 2007, Mindruta et al., 2015). Moreover, participating in accelerator programmes involves costs (Seitz et al., 2019) and whether start-ups actually benefit from them depends on several factors such as the design and morphology of accelerator programmes (Chan et al., 2020; Cohen et al., 2019) and also the characteristics of the founding team (Chatterji et al., 2019). In fact, Hallen et al. (2020) suggest that not all accelerator programmes causally benefit from participating ventures and that some accelerators are better at aiding certain outcomes. It can thus be seen that more empirical research is needed to understand whether and how accelerators assist venture development.

In this paper, we evaluate the benefits of accelerators by comparing the capital structure of accelerated and non-accelerated start-ups. For this purpose we use a novel and worldwide accelerator database, the Entrepreneurship Database Program, which enables us to reduce the confounding effects of survival bias in our results. We only select start-ups which are subject to accelerator programmes during their first 3 years of their life. We empirically analyse three related funding measures, namely: debt-to-capital, external equity-to-capital, and philanthropic-to-capital ratios.

We find that accelerated start-ups have higher external equity ratios than non-accelerated start-ups, after accounting for firm-specific differences and unobserved start-ups factors, particularly when financial conditions deteriorate. We argue that accelerated programmes signal the quality of the start-up for external equity investors and in this way these programmes reduce agency costs. As expected, we confirm that accelerated start-ups raise more funding through philanthropic investors; however, this effect disappears when we control for firm fixed effects. Our findings highlight the role of asymmetric information and the importance of the business cycle in capital structure decisions for start-ups.

Accelerator programmes are a recent phenomenon and their numbers are increasing all over the world, and therefore it is important to understand their benefits and costs and their implications for the capital structure decisions of start-ups. Accordingly, this

[1] See for example, Bates (1990), Brüderl et al., (1992) Cooper et al. (1994), Astebro and Bernhardt (2003), Headd (2003), and Hvide and Moen (2008).

[2] There is still a lack of agreement in the literature regarding whether and how accelerators explain differences in venture performance. For example, Hallen et al. (2020) found that the positive effects of accelerator programmes were not universal, with some even having no impact, and that some accelerator programmes actually cause start-ups to develop more slowly. In addition, the short timeframe of such programmes renders their impact on performance questionable (Cohen 2013; Hochberg 2016). Nevertheless, recent studies such as those of Stayton and Mangematin (2019), Hallen et al. (2020), and Yu (2020) find that being part of an accelerator programme can be beneficial for most ventures.

study offers several contributions to the literature. First, our research corroborates previous studies that identify that accelerator programmes do indeed work (Brown et al., 2019; Hallen et al., 2020). Similar to previous studies, we investigate whether and how these programmes assist the development of ventures, by focusing on how accelerator programmes help start-ups to raise capital. However, we depart from previous research by looking at other sources of funding besides external equity. Participating in an accelerator programme is beneficial for most ventures, as it shortens the time for raising capital and increases the amount of funding from external equity (Hallen et al., 2020; Smith & Hannigan, 2015; Smith et al., 2013). While useful, the previous studies fail to evaluate the effect of accelerated programmes on traditional creditors such as debtholders and philanthropic investors. This lack of attention is striking, considering the role that debt plays in the early life of start-ups. In their initial year of operation, start-ups rely heavily on external debt financing, even start-ups with access to external equity (Robb & Robinson, 2014). Furthermore, start-ups have less financial alternatives available due to capital market frictions, and, therefore, it makes sense to understand the effects of accelerator programmes for a wider variety of funding sources. Contrary to the previous literature, we are able to evaluate whether accelerator start-ups raise more funding from debtholders, external equity investors, and philanthropic investors. These three sources of funding provide us with a better insight into the financing behaviours of different capital providers and the benefits of accelerator programmes for the investors involved. Second, our study highlights the signal mechanism of accelerator programmes to external investors by screening and filtering the better-performing start-ups. In comparison with previous literature, we evaluate whether the signal mechanism is relevant not only for common external equity investors but also for traditional creditors and philanthropic investors. We argue that accelerators are a source of valuable information for external investors and that they reduce information asymmetries and agency problems that can potentially arise between promising start-ups and external investors. By analysing the level of the effect of accelerator programmes on new venture capital structure, we provide empirical evidence of their role in providing information to signal the degree of quality of a company, especially when the economic conditions deteriorate. Importantly, we also highlight the importance of the economic context in shaping the benefits of accelerator programmes. Third, we contribute to the existing literature on capital structure. Despite the wealth of research on the determinants of firm capital structure, it is currently not clear how well the existing literature applies to new ventures, particularly in the cases of accelerated start-ups. Indeed, accelerator programmes have received little attention in the literature on finance (Yu, 2020), where most of the previous research addresses the financial decisions in relation to large and publicly quoted firms. The scarce research on funding decisions and capital structure for accelerated and non-accelerated start-ups is unfortunate, given the role that start-ups play in increasing growth, innovation, and employment, and also the emergence of accelerator programmes in the entrepreneurial landscape. We validate the efficacy of accelerator programmes overall by highlighting their provision of information and their signal role to external equity investors.

This paper is structured as follows. Section 2 reviews the relevant literature on capital structure decisions for start-ups and on the importance of accelerator programmes for new venture development. We also develop the main hypothesis in this section by focusing on the role of accelerator programmes in new venture financing decisions. Section 3 describes the data, the sample, and the relevant variables. In Section 4 we describe the empirical methodology and present the results. Section 5 discusses our findings and the implications for future research. Section 6 concludes and summarises the main results of our study.

2 Literature review

2.1 Capital structure decisions for start-ups

Capital structure decisions regarding start-ups are usually explained by considering the trade-off, pecking order, and agency theories (Degryse et al., 2012; Robb & Robinson, 2014). While the trade-off theory suggests that the tax benefits of debt and bankruptcy and agency costs must be balanced (Modigliani and

Miller, 1958), the pecking order theory (Myers, 1984; Myers & Majluf, 1984) states that start-ups prefer to use internal funds first, followed by debt and external equity, as asymmetric information costs can lead to an increase in the cost of external financing. However, as start-ups grow, entrepreneurs seek external equity to finance their projects (Coleman & Robb, 2011; Vaznyte & Andries, 2019) because these investors can more easily absorb the cash-flow risk (Fluck, 1998). Due to its unlimited life, external equity can spread the moral hazard risk over time. Nevertheless, external equity can also represent a negative signal due to both the lack of incentives by the founders to maximise their profits (Jensen & Meckling, 1976) and also the increase in agency costs, as entrepreneurs tend to give up control of the firm and lose motivation to exert further effort (Ang et al., 2000).

Several other factors might also affect decisions related to the capital structure of start-ups. For example, new firms are more likely to be influenced by their founders and by their preferences and motivations (Cassar, 2004). Furthermore, start-ups have limited operating history, little collateral available to pledge (Cassar, 2004), and face a high degree of uncertainty in terms of demand and resource supply (Denis, 2004). Consequently, they face high levels of asymmetry of information (Berger & Udell, 1998), which prevents them from raising the required amount of capital.Entrepreneurs possess private information regarding their firm's return stream and investment opportunities; however, their ability to objectively convey the merit of their ventures is limited. Additionally, start-ups face moral hazard problems, as investors lack knowledge regarding the actions carried out by the entrepreneurs and they also face adverse selection issues due to the fact that their founders are privy to more information than investors are (Hechavarría et al., 2016). In fact, founders possess a wider knowledge of the potential of their business and a better understanding of its quality than external investors do. Owing to the asymmetry of information, the liquidity constraints theory suggests that capital markets do not provide sufficient funds for profitable ventures (Evans & Jovanovic, 1989).[3] Lenders and external investors are consequently reluctant to lend money to new ventures, as they cannot recognise the quality of the start-up. For example, Brierley (2001) noted that those angel investors and venture capitalists who are potentially willing to finance technology-based firms find it difficult to identify and evaluate the potential of such firms. In this context, accelerator programmes can be used as a mechanism to mitigate these agency costs.

2.2 Accelerators and the capital structure of start-ups

Accelerator programmes are limited-duration programs that provide education, mentorship and network events, and resources for cohorts of start-ups (Cohen & Hochberg, 2014). These programmes affect the capital structure of a start-up by providing and helping to provide funding. They represent one of the key sources of capital for driving innovation and development (Drover et al., 2017).

Accelerated ventures typically agree to receive a small equity investment from the accelerated program. Although the amount is generally smaller when compared with the investments made by early-stage investors (Kerr et al., 2014), by providing such seed capital, to a certain degree accelerators also mitigate the information asymmetry problem, specifically the agency costs associated with moral hazard diminish and a small part of the risk is transferred to the accelerator.

Accelerator programmes also affect the subsequent funding of start-ups. They shorten the time for raising capital and increase the amount of funding from external equity (Hallen et al., 2020; Smith & Hannigan, 2015; Smith et al., 2013; Yu, 2020). For example, Hallen et al. (2020) find that accelerator participants raised more funds during the subsequent 2 to 3 years than almost accepted applicants and also that accelerator programmes increased the speed with which ventures raise their initial round of capital from external equity. Brown et al. (2019) find that accelerated start-ups had easier access to funding. Furthermore, Yu (2020) finds that on average more money is invested in accelerated start-ups.[4] In a certain way, accelerator programmes help start-ups to mitigate the agency costs with external investors

[3] A substantial theoretical literature has emerged which is based on imperfect information and capital market constraint. Two influential theoretical models deserve mentioning: Stiglitz and Weiss (1981) and De Meza and Webb (1987).

[4] In contrast, Seitz et al. (2019) argue that corporate accelerators could have a negative impact on the subsequent funding of start-ups due to lock-in effects and hold-up problems.

and lenders. Their exhaustive selection procedure results in only a few start-ups being accepted (Caley & Kula, 2013; Pauwels et al., 2016). The selection process includes a written application with questions related to the founding team, product idea, and funding status, followed by multiple rounds of interviews with the accelerator partners (Cohen et al., 2019). The acceptance rate for the most selective accelerators is very low and the tendency has been for this rate to decrease, owing to the large number of applicants. Top accelerator programmes are very competitive and typically receive 1000–2000 applications per cohort, of which only 6 to 125 venture applicants are accepted (Hallen et al., 2020). Participating in an accelerator programme can therefore communicate to external investors that the venture is of better quality. Moreover, accelerator programmes evaluate the participating start-ups not only during the selection process but also along the life of the programme and thus obtain an accurate perception regarding the potential and fair value of the start-up (Kim & Wagman, 2016). Accordingly, the participation of a start-up in an accelerator programme sends a signal to the market that the start-up is worth investing in, which thus reduces the gap of information with respect the true value and potential of the entrepreneurial venture between investors and entrepreneurs. Signals[5] are extremely relevant in the entrepreneurial context, as typically information regarding a venture's quality and history is often hard to find (Stuart et al., 1999). When high-quality ventures are associated with high-quality partners, such a partnership sends a credible signal to external investors. For instance, affiliations with higher status venture capital firms serve as effective signals to other investors (Lee et al., 2011). Additionally, accelerator programmes act as brokers, providing mentors for entrepreneurs as well as opportunities to meet with industry experts, peer startups, venture capitalists, and other actors during the acceleration process. These network connections provided by accelerators are also capable of disseminating the quality of the ventures, which thus positively improves the signalling effect. The unique network of accelerator programmes positively contributes to subsequent funding after the completion of the accelerated programme (Christiansen, 2009; Miller & Bound, 2011). Non-accelerator start-ups can also have access to several networks; however, the size of the network is unlikely to be larger than in an accelerator environment, and consequently the credibility of the signal will be lower. Alternatively, non-accelerated start-ups can approach external investors and lenders directly, albeit the screening capabilities of such investors will probably be lower due to the lack of information and the time required to evaluate the quality of these ventures.

Accelerated start-ups are thus regarded to have better quality and to possess greater potential for growth, and consequently they are more likely to raise external equity, debt, and philanthropic capital in comparison to non-accelerated start-ups. To summarise, accelerated programmes have better access to start-up information, which accordingly enables them to estimate the true value of the start-up and consequently to be able to mitigate the agency costs for external equity investors, lenders, and philanthropic investors. Additionally, accelerated programmes also provide access to networks and resources which, in turn, increase their signal effect and the chances of funding from several sources. Therefore, we expect the following:

Hypothesis 1: Accelerated start-ups are more likely to raise a greater proportion of debt in their capital structure than non-accelerated start-ups.

Hypothesis 2: Accelerated start-ups are more likely to use a greater proportion of external equity in their capital structure than non-accelerated start-ups.

Hypothesis 3: Accelerated start-ups are more likely to use a greater proportion of philanthropic capital in their capital structure than non-accelerated start-ups.

The benefits of acceleration programmes and the importance of their signal mechanism could vary according to the market conditions and the business cycle.[6] Firms typically adapt their capital structure according to the business cycle (Baker & Wurgler, 2002) and the same applies to start-ups. Even though no consensus exists regarding the direction of the effect, some components of capital structure are pro-cyclical,

[5] Signals are observable information that is correlated with hard-to-observe attributes (Spence 1973).

[6] Business cycle is defined as the rise and fall of economic growth over a period of time.

while others are counter-cyclical. Previous studies suggest that leverage shows a counter-cyclical pattern, which means that firms tend to issue more debt during contraction periods of the economic cycle and to issue less debt during expansion periods (Al-Zoubi et al., 2018). On the other hand, equity issues are usually pro-cyclical (e.g., Baker & Wurgler, 2002; Bayless & Chaplinsky, 1996). In times of recession, the risk premium increases (Huang & Ritter, 2009), the payment of dividends is reduced, and the agency problem between shareholders and managers increases (Levy & Hennessy, 2007)—all of which negatively affects firms' preference for equity and increases firms' leverage. Choe et al. (1993) reinforced the positive relationship between business cycles and equity issues by suggesting that gains in the stock market reduce the adverse selection problem of firms during boom periods. In addition, internally generated funds and retained earnings increase during expansion periods and conversely decrease during contraction periods, leading to a decrease in leverage in the first case and an increase in the second case (Korajczyk & Levy, 2003). Nevertheless, other studies suggest a different pattern.[7]

Due to their unique characteristics, start-ups face high capital restrictions during contraction periods. The regulation of banks becomes stronger and credit concession for new start-ups becomes harder to come by (Block et al., 2018). Patti and Nigro (2018) found that during economic crises, start-ups tend to use less debt, which could be the result of not only a drop in the supply of credit but also a reduction in demand owing to weaker perspectives for growth. Additionally, the supply of credit by external investors is also scarce and limited during recession periods. Even so, external investors can maximise their gains in these periods by taking advantage of the reduction in the value of start-ups.

[7] For example, Southern (2016) argued that as interest rates are low during economic downturns, firms tend to reduce their leverage in order to improve their financial flexibility. Other studies suggest that during expansion periods, the benefits of debt are greater than the costs and result in pro-cyclical patterns (e.g. Jensen and Meckling 1976; Zwiebel 1996). Moreover, Karabarbounis et al. (2014) found that firms issue more debt during expansion periods, and that the cyclicality of equity depends on how equity is measured. Equity is counter-cyclical if it measured at market value but pro-cyclical if it measured at the book value.

To finance their initial operations, start-ups need to provide signals to the market, particularly those start-ups that are perceived to be of better quality. During economic downturns, investors are more cautious with regard to the future of start-ups (Paik & Woo, 2013) and tend to rely more heavily on signalling effects (Epure and Guasch, 2020). As early-stage firms usually provide less-than-transparent information, these signals are more likely to be used for investors' decisions. Accordingly, during recession periods, start-ups receive larger benefits from participating in acceleration programmes. The acceleration process can be used as a signal for the quality of the start-up and in this way can mitigate the agency cost in a particularly unfavourable scenario. We expect the following:

Hypothesis 4: During recession periods, accelerated start-ups are more likely to use a greater proportion of external investors (debt, external equity, and philanthropic capital) in their capital structure than non-accelerated start-ups.

During expansion periods, start-ups are more likely to generate sufficient revenue to finance their activities and consequently prefer to use internal funds over debt, external equity, and philanthropic capital. This is equally likely for both accelerated and non-accelerated start-ups, as both incur asymmetric information costs. Even though agency costs can be lower in the case of accelerated start-ups, these would not be sufficiently low enough to deter such firms from using internal funds.

3 Data

We use a comprehensive and worldwide accelerator database available from the Entrepreneurship Database Program (EDP) from Emory University in Atlanta, GA, in the USA. The data are supported by the Global Accelerator Learning Initiative, which collects and archives all the data from the EDP's partner accelerators around the world. The accelerators collected data from successful and non-successful applicants between 2016 and 2019 at two distinct time points: (a) at the time of the application and (b) a year after the application, both for successful and unsuccessful applicants. This two-stage data collection represents the most comprehensive year-to-year data on accelerator programmes available from

multiple countries. This dataset excludes those firms which opted not to have their application included in the EDP and also those which applied for more than one accelerator programme. The database includes information on start-up applicants from 170 countries worldwide, surveying more than 23,368 early-stage ventures and 370 accelerator programmes. In relation to firms, the survey includes information regarding the country where the headquarters is located, the number of founders, the number of employees, presence in social media and webpage, the year of foundation, legal status, sector,[8] innovation (copyrights, patents, and trademarks), revenues, whether the start-up had been previously accelerated, and also the sources of financing (own capital, equity, debt, and philanthropic). With regard to entrepreneurs, the survey includes information on age, gender, nationality, degree of education, entrepreneurial and managerial experience, and job experience.

From this database, we select all those for-profit start-ups established after 2014 that applied for an accelerator programme between 2016 and 2019 within their first 3 years of being founded. From this database we construct a panel of start-ups and their capital structure for the years prior to, during, and after successfully and unsuccessfully applying for an acceleration programme. After having excluded start-ups which have missing information regarding their capital structure, we then end up with a sample of 8399 start-ups with information obtained from 16,188 start-year observations.

Table 1 reports the mean characteristics of accelerated and non-accelerated start-ups. We start by computing the capital structure ratios, i.e., debt to capital, external equity to capital, and philanthropic to capital. It should be noted that total capital includes the total amount of funding raised by the start-up in each year, namely, capital provided by the founders (owned equity); capital provided by business angels, venture capitalists, government agencies, and other firms (external equity); debt provided by banks, financial institutions, government agencies, other companies, family members, employees, and other individuals (debt); capital provided by government agencies, foundations or other nonprofits institutions, accelerators or fellowship programmes, business plan competitions, crowdfunding campaigns, friends or families members, and other sources (philanthropic capital).

The ratios of debt-to-total capital—which is defined as a percentage of debt to total capital—is 8% and 7% for accelerated and non-accelerated start-ups, respectively. A sample mean test rejects the null hypothesis that debt to capital in accelerated and non-accelerated start-ups is the same at the 5% level. In addition, the external equity-to-capital and philanthropic-to-capital ratios are different between accelerated and non-accelerated start-ups, with these differences being statistically significant at the 1% level for both cases. The average ratios of external equity to capital are 19.2% for accelerated start-ups and 12.7% for non-accelerated start-ups. As expected, the average philanthropic-to-capital ratios are 19.1% and 12.8% in accelerated and non-accelerated firms, respectively. The philanthropic ratio is larger for accelerated start-ups, as accelerators first appeared to overcome start-up need for financial capital (Hallen et al., 2020; Hoffman & Radojevich-Kelley, 2012). In fact, Christiansen (2009) and Miller and Bound (2011) considered access to financial capital to be the most appealing benefit of accelerators.

In terms of the characteristics of the start-ups, we observe that, on average, accelerated start-ups are slightly younger (1.36 years in accelerated start-ups versus 1.44 years in non-accelerated start-ups) and have greater sales revenue (€809,458 in accelerated start-ups versus €152,155 in non-accelerated start-ups) and more full-time employees (321 in accelerated start-ups versus 231 in non-accelerated firms). Furthermore accelerated start-ups are less profitable (52% of accelerated start-ups have positive margin versus 57% in non-accelerated start-ups). It should be noted, however, that the differences in sales revenue and full-time employees are not statistically significant.

On average, accelerated start-ups are established by a slightly larger number of founders (2.29 in accelerated start-ups versus 2.24 in non-accelerated start-ups). With regard to the characteristics of the founders, we observe that, on average, founders of accelerated start-ups are slightly younger (34 years old in accelerated companies versus 35 years old in non-accelerated companies), are less male (69.8% of

[8] The main sectors are agriculture, handicraft, culture, education, energy, environment, financial services, health, housing development, information and communication technology, infrastructure/facility development, supply chain services, technical assistance services, tourism, water, and others.

Table 1 Descriptive statistics

	Accelerated	Non-accelerated	Difference
Capital structure			
Debt-to-capital			
Mean	0.084	0.072	−0.012**
SD	0.225	0.214	
External equity-to-capital			
Mean	0.192	0.128	−0.065***
SD	0.331	0.285	
Philantropic-to-capital			
Mean	0.191	0.128	−0.062***
SD	0.32	0.2836	
Firm Characteristics			
Firm's age			
Mean	1.362	1.436	0.073***
SD	1.118	1.11	
Sales			
Mean	809,458	152,155	−657,303
SD	43,800,000	2,962,967	
Employees			
Mean	321.301	231.681	-89.62
SD	21,819.43	16,854.54	
Positive margin			
Mean	0.521	0.568	0.048***
SD	0.5	0.495	
Patent			
Mean	0.168	0.139	−0.029***
SD	0.374	0.346	
Intellectual property			
Mean	0.483	0.433	−0.05***
SD	0.5	0.496	
Invention based			
Mean	0.650	0.598	−0.053***
SD	0.477	0.490	
Website			
Mean	0.829	0.765	−0.064***
SD	0.376	0.424	
Social media			
Mean	0.835	0.762	−0.073***
SD	0.372	0.426	
No. of founders			
Mean	2.294	2.241	−0.054***
SD	0.73	0.751	
Founders characteristics			
Founder age			
Mean	33.836	34.532	0.697***
SD	8.225	9.517	
Male founders			

Table 1 (continued)

	Accelerated	Non-accelerated	Difference
Mean	0.698	0.711	0.013**
SD	0.358	0.351	
Foreign founders			
Mean	0.18	0.142	−0.037***
SD	0.330	0.305	
High-educated founders			
Mean	0.847	0.830	−0.017***
SD	0.310	0.327	
Managerial experience			
Mean	0.664	0.639	−0.024***
SD	0.473	0.480	
Entrepreneurial experience			
Mean	0.618	0.597	−0.022***
SD	0.486	0.491	

This table reports the mean and standard deviation for several characteristics for accelerated and non-accelerated start-ups. Debt to capital is the ratio between the debt and total capital of the start-up. External equity to capital is the ratio between the external equity and the total capital of the firm. Philanthropic to capital is the ratio between the philanthropic financing and the total capital of the start-up. Firm's age is the number of years of the start-up since its founding. Sales is the value of revenues of the start-up since founding. Full-time employees is the number of employees working full time in the start-up. Positive margin is the dummy variable equal to 1 if a start-up has a positive profit margin and 0 otherwise. No. of founders is the number of founders of the start-up. Founders' age is the mean age of the founders of the start-up. Male founders is the ratio between the number of male founders and the total number of founders. Foreign founders is the ratio between the number of foreign founders and the total number of founders. High-educated founders is the ratio between the number of founders who have at least a bachelor's degree and the total number of founders. Managerial experience is the ratio between the number of founders with previous managerial experience and the total number of founders. Entrepreneurial experience is the ratio between the number of founders with previous entrepreneurial experience and the total number of founders. Website is a dummy variable which is equal to 1 if the start-up has a website and 0 otherwise. Social Media is a dummy variable which is equal to 1 if the start-up uses Facebook, Twitter, or LinkedIn, and 0 otherwise. Patent is a dummy variable equal to 1 if the start-up has reported any patent and 0 otherwise. Intellectual property is a dummy variable equal to 1 if the start-up has copyrights or trademarks and 0 otherwise. Invention based is a dummy variable equal to 1 if the start-up builds upon newly created technology and 0 otherwise

male founders in accelerated start-ups versus 71.1% on non-accelerated start-ups), are more likely to be foreign (18% of foreign founders in accelerated firms versus 14% in non-accelerated firms), are more educated (85% of the founders have, at least, a bachelor degree in accelerated start-ups versus 83% in non-accelerated firms), and are more likely to have managerial experience (66.4% of founders have managerial experience in accelerated firms versus 63.9% in non-accelerated start-ups) and entrepreneurial experience (62% of founders have entrepreneurial experience in accelerated start-ups versus 60% in non-accelerated companies). All these differences are statistically significant at the 1% level, except for the case of gender, which is only significant at the 5% level.

In terms of other characteristics, we observe that 83% of the accelerated start-ups have a website compared to 76% of non-accelerated start-ups; 83% of the accelerated start-ups are present on one of social media platforms (LinkedIn, Facebook, or Twitter), while this percentage reduces to 76% in non-accelerated companies; 17% of accelerated start-ups have patents compared to 14% of non-accelerated start-ups. A total of 48% of accelerated start-ups own intellectual property rights (copyrights or trademarks), although this number decreases to 43% in non-accelerated companies; in addition, 65% of accelerated firms are invention based, while in the case of non-accelerated companies, only 60% are invention based. The differences between accelerated and non-accelerated start-ups are statistically significant at the 1%

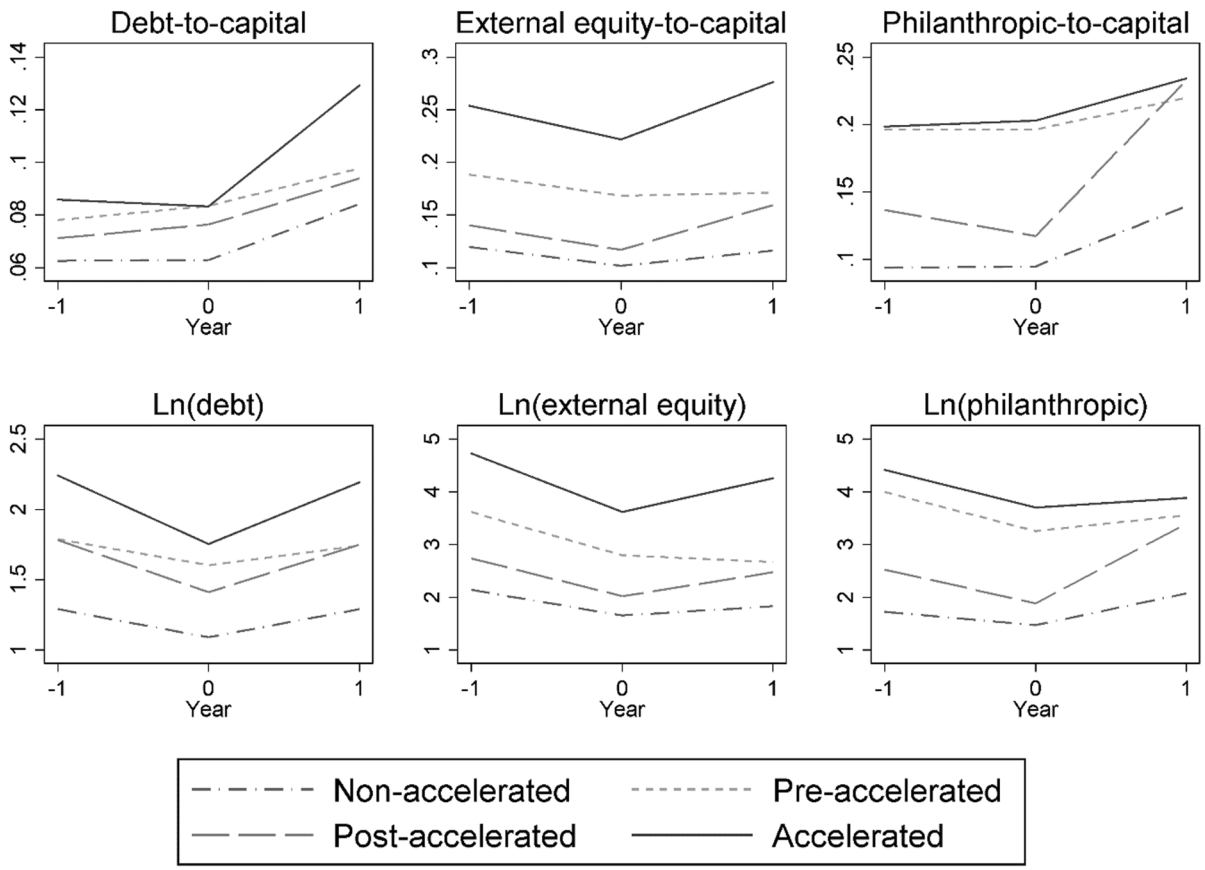


Fig. 1 The capital structure of start-ups before and after being accelerated. Note: The capital structure of start-ups before and after being accelerated. These graphics illustrate the evolution of the debt-to-capital, external equity-to-capital, and philanthropic-to-capital ratios, and also the total amounts of debt, external equity, and philanthropic capital for four groups of start-ups, namely, start-ups which have never been accelerated (Non-accelerated); start-ups which were previously accelerated but were rejected in the current accelerated programme (Pre-accelerated); start-ups which were not previously accelerated but were accepted in the current accelerated programmes (Post-accelerated), and; start-ups which were accelerated through the entire period under study (Accelerated)

level for all these characteristics. In order to examine whether these factors jointly affect the capital structure of start-ups, we implement a multivariate regression analysis in the next section.

We observe interesting developments in the capital structure with regard to start-ups which transition from being accelerated to non-accelerated, or changed from non-accelerated to accelerated status, and also for start-ups which continue to be either accelerated or non-accelerated over the entire period under study. Figure 1 shows the capital structure of start-ups divided into four groups: non-accelerated, start-ups which have never been accelerated; pre-accelerated, start-ups which were previously accelerated but were rejected during the following accelerated programme; post-accelerated, start-ups which were not previously accelerated, but were accepted during the following accelerated programmes; and, finally, accelerated, start-ups which were previously accelerated and were accepted for the following acceleration programme.

Starting with the debt-to-capital ratio, on the left-hand side and top row of Fig. 1, we can observe that non-accelerated start-ups have the lowest ratio, whereas accelerated start-ups have the highest ratio of debt to capital. Nevertheless, all start-ups experience an increase in the debt-to-capital ratio after applying for an accelerated programme, independently of being accepted or not being accepted for the

programme; however, this increase is larger for accelerated and post-accelerated start-ups. Similarly, the external equity-to-capital ratio, which is shown in the middle of the graph, presents the largest increase for accelerated and post-accelerated start-ups, while the external equity-to-capital ratio of non-accelerated and pre-accelerated start-ups remains mostly the same. Once again, non-accelerated start-ups tend to have the lowest ratio, while accelerated start-ups rely more on external equity. In terms of philanthropic-to-capital ratio, which is shown in the right-hand-side graph, we observe that non-accelerated start-ups have the lowest ratio, while accelerated start-ups present the highest ratio. All start-ups experience an increase in the philanthropic-to-capital ratio after applying for an accelerated programme. Nevertheless, post-accelerated start-ups were seen to experience a stronger increase. In the second row of the graph, we plot the logarithm values of debt, external equity, and philanthropic capital. A similar trend is observed for the logarithm values, whereby all start-ups experience an increase in debt and philanthropic capital, even though the increase is greater for accelerated and post-accelerated start-ups. In terms of external equity, only accelerated and post-accelerated start-ups were observed to experience an increase, while for non-accelerated start-ups and pre-accelerated start-ups, the value remains the same or decreases, respectively.

4 Results

To control for unobserved factors, we estimate the following least square regression specification:

$$\text{Capital Structure}_{it} = \beta_t + \beta_i + \beta_1 \text{ Accelerated}_{it} + X'_{it}\beta_2 + \epsilon_{it} \quad (1)$$

where Accelerated is an indicator variable that takes 1 in a given year if a start-up was selected for an accelerator programme and 0 otherwise. X_{it} is a set of time-variant firm control variables: profit margin, a dummy variable which equals 1 if the start-up has a positive profit margin and 0 otherwise; firm age is measured by logarithm of the number of years of the start-up; and size is measured by the logarithm of firm sales. These factors have been found to be the major determinants of firm capital structure (e.g. Hovakimian et al., 2001; Rajan & Zingales, 1995).

We use three dependent variables, namely, debt over total capital (debt-to-capital ratio), external equity capital over total capital (external equity-to-capital ratio), and philanthropic capital over total capital (philanthropic-to-capital ratio). To control for time-invariant firm heterogeneity and the macroeconomic environment, we include firm (θ_i) and time fixed (λ_t) effects.

Table 2 reports the results of the pooled OLS regressions for the three dependent variables. In these specifications we include additional time-constant control variables to account for founder and start-ups characteristics. We control for start-ups growth opportunities by including three measures: patent, a dummy variable if the start-up has any patents; intellectual property, a dummy variable if the start-up has copyrights or trademarks; and invention based model, a dummy variable if the start-up is built upon newly created technology. To control for digital/e-commerce start-ups, we include two variables: website, which is measured by a dummy variable if the start-up has a website, and social media, which is measured by a dummy variable if the start-up is present in one of the social media platforms (Facebook, Twitter, or LinkedIn). As suggested in the previous literature, founders play an important role in the choice of the funding of start-ups (Cassar, 2004). Accordingly, we include a set of time-constant characteristics for the founder, namely, number of founders; the average age of the founders; percentage of male founders; percentage of foreign founders; percentage of high educated founders (with bachelor degree or higher); managerial experience, which is measured by a dummy variable if one of the founders previously worked as a CEO/executive director or as a senior director; and also entrepreneurial experience, which is measured by a dummy variable if one of the founders previously established a for-profit start-up. We also include year, accelerated programme, country, and sector fixed effects.

In model 1, the coefficient associated with the dummy variable for programme acceleration is not statistically significant, which suggests that the proportion of debt in the capital structure is not different between accelerated and non-accelerated start-ups. We did not find support for Hypothesis 1. However, the coefficients associated with programme acceleration in models 2 and 3 are

Table 2 Pooled OLS: capital structure for accelerated and non-accelerated start-ups

	Debt to capital	External equity to capital	Philantropic to capital
	(1)	(2)	(3)
Accelerated	0.003	0.040***	0.061***
	(0.004)	(0.005)	(0.005)
Firm characteristics			
Firm's age	0.022***	0.022***	0.025***
	(0.006)	(0.008)	(0.008)
Sales	0.007***	0.007***	−0.002***
	(0.000)	(0.001)	(0.001)
Positive margin	−0.026***	−0.073***	0.042***
	(0.005)	(0.006)	(0.006)
Patent	−0.005	0.024**	0.016*
	(0.007)	(0.009)	(0.009)
Intellectual property	0.002	0.031***	−0.011*
	(0.005)	(0.006)	(0.006)
Invention based	−0.011**	0.024***	0.029***
	(0.005)	(0.006)	(0.006)
Website	−0.013**	0.000	0.023***
	(0.006)	(0.006)	(0.009)
Social media	-0.004	0.028***	0.007
	(0.006)	(0.007)	(0.007)
No. of founders	-0.005	0.031***	0.028***
	(0.006)	(0.008)	(0.008)
Founder characteristics			
Founders' age	0.018*	−0.082***	−0.147***
	(0.010)	(0.013)	(0.014)
Male founders	0.010	0.049***	−0.054***
	(0.007)	(0.008)	(0.010)
Foreign founders	−0.004	0.036***	−0.004
	(0.008)	(0.012)	(0.011)
High-educated founders	−0.004	0.000	0.005
	(0.008)	(0.010)	(0.011)
Managerial experience	0.003	0.018***	−0.033***
	(0.005)	(0.007)	(0.007)
Entrepreneurial experience	0.008*	−0.003	−0.009
	(0.005)	(0.006)	(0.007)
Constant	−0.150***	0.312***	0.695***
	(0.058)	(0.083)	(0.067)
Year fixed effects	Yes	Yes	Yes
Programme fixed effects	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes

Table 2 (continued)

	Debt to capital	External equity to capital	Philantropic to capital
Sector fixed effects	Yes	Yes	Yes
Observations	16,188	16,188	16,188
R-squared	0.113	0.193	0.142

This table evaluates the impact of programme acceleration on the capital structure of start-ups. The dependent variables are debt to capital, external equity to capital, and philanthropic to capita. Debt to capital is the ratio between the debt and total capital of the start-up. External equity to capital is the ratio between the external equity and the total capital of the firm. Philanthropic to capital is the ratio between the philanthropic financing and the total capital of the start-up. Firm's age is the number of years of the start-up since its founding. Sales is the logarithm of start-up sales. Positive margin is the dummy variable equal to 1 if a start-up has a positive profit margin and 0 otherwise. Patent is a dummy variable equal to 1 if the start-up has reported any patent and 0 otherwise. Intellectual property is a dummy variable equal to 1 if the start-up has copyrights or trademarks and 0 otherwise. Invention based is a dummy variable equal to 1 if the start-up builds upon a newly created technology. Website is a dummy variable which is equal to 1 if the start-up has a website and 0 otherwise. Social Media is a dummy variable which is equal to 1 if the start-up uses Facebook, Twitter, or LinkedIn, and 0 otherwise. No. of founders is the number of founders of the start-up. Founders' age is the mean age of the founders of the start-up. Male founders is the ratio between the number of male founders and the total number of founders. Foreign founders is the ratio between the number of foreign founders and the total number of founders. High-educated founders is the ratio between the number of founders who have at least a bachelor's degree and the total number of founders. Managerial experience is the ratio between the number of founders with previous managerial experience and the total number of founders. Entrepreneurial experience is the ratio between the number of founders with previous entrepreneurial experience and the total number of founders. Standard errors in brackets are clustered at the firm level. All regressions include year, accelerator programme, country, and sector fixed effects. *** denotes 1% significant level, ** denotes 5% significant level, and * denotes 10% significant level

positive and statistically significant at the 1% level for external equity and philanthropic-to-capital ratios, which suggests that accelerated start-ups are able to use more external and philanthropic capital in their capital structure after controlling for firm and founder characteristics, year, type of accelerated programme, country, and sector fixed effects. The differences between accelerated and non-accelerated start-ups in external equity and philanthropic-to-capital ratios are 4% and 6%, respectively. This finding is consistent with Hypotheses 2 and 3, which suggest that accelerated start-ups are less affected by asymmetric information costs because these firms already went through a selection and monitoring process when they applied for the acceleration programme. More information is typically available regarding accelerated start-ups, which implies that these start-ups benefit from easier access to external equity and philanthropic capital in comparison to non-accelerated ones.

In terms of control variables, all coefficients are statistically significant at the 1% level. Firm age positively affects all the ratios (debt-to-capital, external equity-to-capital, and philanthropic-to-capital ratios) and, interestingly, older start-ups tend to have more access to finance. On the whole, start-ups prefer to finance themselves with retained earnings first and then with external financing at a later stage (Huynh et al., 2018). Start-up size positively affects debt and external equity-to-capital ratios, where a 1% increase in sales is associated with a 0.7% increase both in debt and external equity-to-capital ratios. Information asymmetries are typically less severe in larger than in smaller start-ups due to the fact that more information is available regarding larger start-ups and also that the likelihood of default is also lower. In contrast, larger start-ups are less likely to raise funds from business plan competitions and crowdfunding campaigns. As start-ups grow, they increasingly need more funding to leverage their projects (Cassar & Holmes, 2003) and the weight of the capital raised through philanthropic investors decreases. Profit margin is negatively correlated with debt-to-capital and external equity-to-capital ratios, with these negative correlations being explained by firms' preference

for internal, rather than external funding, which is line with the pecking-order theory (Brealey et al., 1977). Nevertheless, this effect differs in the case of the philanthropic-to-capital ratio, as profit margin is positively correlated with philanthropic-to-capital ratio, which suggests that philanthropic investors are more likely to provide funds when the business model is profitable and scalable. The estimate for the invention-based dummy is negative and is statistically significant for debt-to-capital but is positive for the external equity-to-capital and philanthropic-to-capital ratios. This result is consistent with the previous literature, which suggests that the expected costs of financial distress are larger for start-ups with better growth opportunities and therefore the relationship between leverage and proxies for growth opportunities should be negative. As expected, external equity and philanthropic investors fill the funding gap for start-ups which possess better growth opportunities. Start-ups with patents are positively correlated with external equity and philanthropic investors. However, the effects of intellectual property rights are less obvious, whereby start-ups with intellectual property—copyrights or trademarks but excluding patents—have proportionally more external equity in their capital structure, but less philanthropic capital. By the same token, the coefficient associated with start-ups which have a website or are present on at least one social media platform is negative and statistically significant for debt-to-capital but positive for the external equity-to-capital and philanthropic-to-capital ratios. The negative correlation with leverage can be explained by the increases in information asymmetry in the case of digital/e-commerce start-ups.

In terms of founder characteristics, founders' age seems to be positively correlated with debt ratio and are negatively correlated with external equity and philanthropic ratios. As suggested by Sanyal and Mann (2010), older founders tend to have a better access to credit market. Male, foreign founders with managerial experience are positively correlated with

Table 3 Fixed effects: capital structure for accelerated and non-accelerated start-ups

	Debt to capital	External equity to capital	Philanthrophic to capital
	(1)	(2)	(3)
Accelerated	0.001	0.008*	0.003
	(0.003)	(0.004)	(0.005)
Firm's age	−0.032***	−0.069***	−0.054***
	(0.012)	(0.014)	(0.016)
Sales	0.003***	0.005***	0.004***
	(0.001)	(0.001)	(0.001)
Positive margin	−0.012	−0.034**	0.049***
	(0.012)	(0.015)	(0.017)
Constant	0.026**	0.102***	0.032*
	(0.013)	(0.015)	(0.018)
Year effects	Yes	Yes	Yes
Firm effects	Yes	Yes	Yes
Observations	16,188	16,188	16,188
No. of start-ups	8399	8399	8399
Within R2	0.011	0.012	0.012
σ_u	0.198	0.279	0.276
σ_e	0.131	0.165	0.179
σ	0.237	0.324	0.329
ρ	0.694	0.740	0.704

This table evaluates the impact of programme acceleration on the capital structure of start-ups. The dependent variables are debt to capital, external equity to capital, and philanthropic to capital. Debt to capital is the ratio between the debt and total capital of the start-up. External equity to capital is the ratio between the external equity and the total capital of the firm. Philanthropic to capital is the ratio between the philanthropic financing and the total capital of the start-up. Firm's age is the number of years of the start-up since its founding. Sales is the logarithm of start-up sales. Positive margin is the dummy variable equal to 1 if a start-up has a positive profit margin and 0 otherwise. Standard errors in brackets are clustered at the firm level. All models include year and firm fixed effects. The firm-level variance is σ_u, idiosyncratic variance isσ_e, total variance isσ, and ρ is the ratio of firm to total variance. *** denotes 1% significant level, ** denotes 5% significant level, and * denotes 10% significant level

external equity ratios but are negatively correlated with philanthropic capital ratios, which could be explained by a preference for risk aversion or credit discrimination (Coleman & Cohn, 2000). Finally, founders with entrepreneurial experience present a higher debt ratio.

We next estimate the same regressions as in Table 2, but this time including firm fixed effects. Unobserved heterogeneity at the firm level plays an important role in explaining capital structure ratios (e.g. Lemmon et al., 2018), and it should be noted that the acceleration decision is strongly correlated with firm-specific unobservable factors and does not include those factors that could bias the results. Firm fixed effects account for the time-constant unobserved characteristics of firms. The results are reported in Table 3.

In column 1 of Table 3, the coefficient for acceleration is not statistically significant, which suggests that the proportion of debt in the capital structure of accelerated and non-accelerated start-ups does not differ between these start-ups. Similarly, the coefficient for the philanthropic-to-capital ratio is positive but is not statistically significant (see column 3 of Table 3). Nevertheless, our results suggest that being accelerated can have meaningful effects for the external equity-to-capital ratio. The estimate regarding acceleration status for the external equity-to-capital ratio is 0.8%, compared to 4%, when firm fixed effects are controlled for (see column 2 of Table 3). These results highlight that differences in the capital structure between accelerated and non-accelerated start-ups are smaller within firms than across firms which never change their acceleration status. The estimates of 4% for external equity to capital represents the upper boundary, which is derived from the cross-section variation between accelerated and non-accelerated start-ups, while the estimate for 0.8% is the lower boundary, which is driven by start-ups which change their status from accelerated to non-accelerated, and vice-versa.

Table 4 Logistic regression: determinants of being selected for an acceleration programme

	Accelerated
	(1)
Firm characteristics	
Firm's age	0.173***
	(0.053)
Sales	0.046***
	(0.005)
Positive margin	−0.192***
	(0.042)
Patent	0.158***
	(0.058)
Intellectual property	0.055
	(0.043)
Invention based	0.245***
	(0.046)
Website	0.303***
	(0.066)
Social media	0.258***
	(0.055)
No. of founders	0.199***
	(0.054)
Founder characteristics	
Founders' age	−0.537***
	(0.096)
Male founders	−0.260***
	(0.063)
Foreign founders	0.131*
	(0.071)
High-educated founders	0.264***
	(0.076)
Managerial experience	0.080*
	(0.048)
Entrepreneurial experience	0.080*
	(0.046)
Constant	−0.424
	(0.541)
Year effects	Yes
Programme effects	Yes
Country effects	Yes
Sector effects	Yes

Table 4 (continued)

	Accelerated
Observations	15,997
Log Likelihood	−9694

This table evaluates the determinants of being selected for an accelerated programme. The dependent variable is dummy variable equal to 1 if in a given year a start-up was selected and finished an accelerator programme and 0 otherwise. Firm's age is the number of years of the start-up since its founding. Sales is the logarithm of start-up sales. Positive margin is the dummy variable equal to 1 if a start-up has a positive profit margin and 0 otherwise. Patent is a dummy variable equal to 1 if the start-up has reported any patent and 0 otherwise. Intellectual property is a dummy variable equal to 1 if the start-up has copyrights or trademarks and 0 otherwise. Invention based is a dummy variable equal to 1 if the start-up builds upon newly created technology and 0 otherwise. Website is a dummy variable which is equal to 1 if the start-up has a website and 0 otherwise. Social Media is a dummy variable which is equal to 1 if the start-up has Facebook, Twitter, or LinkedIn, and 0 otherwise. No. of founders is the number of founders of the start-up. Founders age is the mean age of the founders of the start-up. Male founders is the ratio between the number of male founders and the total number of founders. Foreign founders is the ratio between the number of foreign founders and the total number of founders. High-educated founders is the ratio between the number of founders who have at least a bachelor's degree and the total number of founders. Managerial experience is the ratio between the number of founders with previous managerial experience and the total number of founders. Entrepreneurial experience is the ratio between the number of founders with previous entrepreneurial experience and the total number of founders. Standard errors in brackets are clustered at the firm level. The regressions includes year, programme accelerator, country, and sector fixed effects. *** denotes 1% significant level, ** denotes 5% significant level, and * denotes 10% significant level

In terms of control variables, the age of start-ups is statistically significant and is negatively correlated with debt, external equity, and philanthropic ratios. In contrast, sales seem to be positively and statistically correlated with the same three ratios. Profit margin is statistically significant for philanthropic ratio but is negatively correlated with external equity ratio.

A limitation of the above-mentioned analysis is that it fails to take into consideration the fact that the decision of being accelerated can be endogenous and neither does it rule out the sorting effect between acceleration and the ventures themselves. In fact, the decision to apply and the result of being selected for an accelerated programme depend on various start-ups and founder characteristics. We evaluate the main

Table 5 Propensity score matching

	Debt to capital	External equity to capital	Philanthropic to capital
All years	0.002	0.043***	0.063***
	(0.004)	(0.006)	(0.006)
2015	0.021	0.041*	0.059***
	(0.018)	(0.025)	(0.022)
2016	0.011	0.017	0.090***
	(0.010)	(0.015)	(0.015)
2017	−0.008	0.047***	0.079***
	(0.009)	(0.012)	(0.012)
2018	−0.010	0.060***	0.070***
	(0.010)	(0.013)	(0.013)
2019	0.002	0.062***	0.033*
	(0.017)	(0.023)	(0.021)

Debt to capital is the ratio between the debt and total capital of the start-up. External equity to capital is the ratio between the external equity and the total capital of the firm. Philanthropic to capital is the ratio between the philanthropic financing and the total capital of the start-up. The average treatment effect on the treated (ATT) are calculated using kernel matching. Bootstrapped standard errors are reported in the brackets. *** denotes 1% significant level, ** denotes 5% significant level, and * denotes 10% significant level

determinants for being selected for an acceleration programme in Table 4 using logit regression. The dependent variable is a dummy which equals 1 if in a given year a start-up was selected for an accelerator programme or 0 otherwise. We control for both various start-up characteristics (firm's age, sales, profit margin, number of founders, website, social media, patents, intellectual property, and invention based) and founders' characteristics (gender, age, education, nationality, and managerial and entrepreneurial experience), and we also include year, accelerated programme, country, and sector fixed effects.

Based on Table 4, we can observe that accelerated and non-accelerated start-ups differ substantially. Indeed, accelerated start-ups are significantly older, larger, and less profitable, with better growth opportunities (evaluated in terms of patents and invention-based business model), and are more digital based (measured in terms of social media and website) than non-accelerated start-ups. The characteristics of accelerated and non-accelerated founders are also different. For example, accelerated founders are younger, less likely to be male, more likely to be foreign, more educated, and possess

more entrepreneurial and managerial experience. With such a pronounced difference between accelerated and non-accelerated start-ups, we run the risk of estimating the condition distribution of start-up capital structure in terms of characteristics which are nonexistent for non-accelerated start-ups.

Ideally, to carry out a comparison between the capital structure of accelerated and non-accelerated start-ups, we should compare identical firms in terms of all the characteristics, except for the accelerated status. For this purpose, we employ the propensity score matching method (Heckman et al., 1997), which is an approach that estimates the predicted probabilities of being accelerated (treatment) versus non-accelerated (control) groups, based on observed characteristics. We use the set of start-ups and founder characteristics to generate the probability of being accelerated (the propensity score). Using the propensity scores, we match each accelerated start-up score with a set of non-accelerated start-ups which have similar propensity scores. To perform the matching, we rely on a kernel estimator which uses the weighted average of all non-accelerated start-ups with more weighting given to non-accelerated start-ups whose propensity scores are closer to those of the accelerated start-ups. The weightings are inversely proportional to the distance between the propensity scores of both the treated and control groups.

A basic requirement for using the matching method is a sufficiently large overlap between the propensity scores of accelerated and non-accelerated start-ups, in order to validate the common support condition. Put simply, this method ensures that any combination of characteristics observed in the treatment group can also be observed in the control group. The propensity score distributions between accelerated and non-accelerated start-ups present a substantial overlap between both groups of start-ups (see Appendix Fig. 2).

Table 5 reports the average treatment effect on treated (ATT) of our main variables of interest. The ATT estimates the counterfactual view of what an accelerated start-up's capital structure would be if the start-ups were non-accelerated. For this purpose, a comparison is carried out between the capital structure of those accelerated start-ups which have the same set of characteristics as those of non-accelerated start-ups.

Starting with debt to capital, we note that after the matching, the results continue to show that the differences between accelerated and non-accelerated start-ups are not sizeable for all years, neither individually for each year. In contrast, accelerated start-ups have higher external capital ratios than non-accelerated start-ups, ranging from 4.1% in 2015 to 6.2% in 2019. These differences are not significant in 2016. Compared to the 4% external equity-to-capital ratio difference between accelerated and non-accelerated start-ups reported in column 2 of Table 3, the matching procedure produces external equity ratio differences of 4.3% for all the years, which suggests that the differences in external capital funding do not cease to exist, even when we compare similar accelerated and non-accelerated start-ups. For the philanthropic-to-capital ratio, we also confirm that accelerated start-ups have a higher incidence of capital which is raised from the government, business competitions, accelerator programmes, crowdfunding, and other sources after comparing accelerated and non-accelerated start-ups with similar characteristics. The difference ranges from 3.3% in 2019 to 9% in 2016.

Lastly, we explore the effect of business cycle conditions on the capital structures of accelerated and non-accelerated start-ups. We use country-economic shocks to proxy for the state of the economic environment in which the start-up is located. For each start-up, the data provide information regarding in which country the firm's headquarters is located. These economic shocks are useful to proxy for credit availability. During economic downturns, start-ups may experience difficulty in raising debt due to a decline in the collateral values of their assets. For this purpose, we transform Eq. 1 to enable the acceleration dummy to be able to vary with the state of the economy where the start-up is located, as follows:

$$\text{Capital Structure}_{it} = \beta_t + \beta_i + \rho^L \times F(z_{i,t})\text{Accelerated}_{it} + \rho^H \times (1 - F(z_{i,t})) + X'_{it}\beta_2 + \epsilon_{it} \quad (2)$$

with $F(z_{it}) = \frac{\exp(-\gamma z_{it})}{1+\exp(-\gamma z_{it})}, \gamma > 0$, in which z_{it} is an indicator for the state of the economy (the real GDP growth), which is normalised to have a zero mean and unit variance. The weightings assigned to each regime vary between 0 and 1, according to the weighting function $F(.)$, and thus $F(z_{it})$ can be interpreted as the probability of being in a given state of the economy. The ρ^L and ρ^H coefficients capture the impact of the acceleration being based on the start-up's capital structure in cases of extreme recessions ($F(z_{it}) \approx 1$ when z goes to

Table 6 Economic state contingent fixed effects

	Debt to capital	External equity to capital	Philanthropic to capital
	(1)	(2)	(3)
Accelerated*recession	0.004	0.020*	0.010
	(0.008)	(0.012)	(0.014)
Accelerated*expansion	−0.002	−0.005	−0.003
	(0.008)	(0.010)	(0.013)
Firm's age	−0.033***	−0.070***	−0.053***
	(0.012)	(0.015)	(0.016)
Sales	0.003***	0.005***	0.004***
	(0.001)	(0.001)	(0.001)
Positive margin	−0.012	−0.034**	0.049***
	(0.012)	(0.015)	(0.017)
Constant	0.025*	0.101***	0.033*
	(0.013)	(0.015)	(0.018)
Year effects	Yes	Yes	Yes
Firm effects	Yes	Yes	Yes
Observations	16,096	16,096	16,096
R-squared	0.012	0.013	0.012
No. of start-ups	8354	8354	8354
Within R2	0.012	0.013	0.012
σ_u	0.198	0.279	0.276
σ_e	0.131	0.165	0.179
σ	0.237	0.324	0.330
ρ	0.693	0.740	0.704

The table evaluates the impact of programme acceleration on start-ups capital structure. The dependent variables are debt to capital, external equity to capital, and philanthropic to capital. Debt to capital is the ratio between the debt and total capital of the start-up. External equity to capital is the ratio between the external equity and the total capital of the firm. Philanthropic to capital is the ratio between the philanthropic financing and the total capital of the start-up. Firm's age is the number of years of the start-up since its founding. Sales is the logarithm of start-up sales. Positive margin is the dummy variable equal to 1 if a start-up has a positive profit margin and 0 otherwise. Standard errors in brackets are clustered at the firm level. All models include year and firm fixed effects. The firm-level variance is σ_u, idiosyncratic variance isσ_e, total variance isσ, and ρ is the ratio of firm to total variance *** denotes 1% significant level, ** denotes 5% significant level, and * denotes 10% significant level

minus infinity) and booms ($1 - F(z_{it}) \approx 1$ when z goes to plus infinity), respectively.[9][10] This approach is inspired by the smooth transition autoregressive model developed by Granger and Terasvirta (1993).

Table 6 shows the results of estimating the state contingent Eq. 2, using a fixed effects estimator. In columns 1–3 of Table 6, the coefficients for interest are the interaction coefficients between accelerated and recession, and accelerated and expansion. We are able to observe that when the business cycle conditions are favorable, our results are not statistically significant, which suggests that there is no difference between the capital structure of accelerated and non-accelerated start-ups.

On the other hand, when the state of the economy deteriorates, our results suggest an increase in external equity-to-capital ratios, which is statistically significant at the 10% level, and a consequential reduction of 3.4% in own equity-to-capital ratio. This result is consistent with Hypothesis 4 and suggests that accelerated start-ups opt to procure external equity investors during periods of recession, which is when the cost of information asymmetry is relatively higher. It should be highlighted that accelerated start-ups are less affected by asymmetric information

[9] $F(z_{it})$= 0.5 is the cutoff between weak and strong economic activity.

[10] We choose $\gamma = 1.5$, following Auerbach and Gorodnichenko (2012), whereby the economy spends about 20% of the time in a recessionary regime—which is defined as $F(z_{it}) > 0.8$. Our results hardly change when using alternative values of the γ parameter between 1 and 6.

costs because they are perceived to possess higher quality. This result is consistent with supply-side explanations, whereby investors become more risk averse and shy away from financing firms which are subject to relatively high asymmetric information (e.g., Caballero & Krishnamurthy, 2008). Accelerated start-ups notably rely more on external equity than on debt financing. With regard to the percentage of philanthropic capital and debt during downturn periods, our results are not statistically significant, which is contrary to our expectations.

In terms of the control variables, start-up age is statistically significant and is negatively correlated with debt, external equity, and philanthropic ratios. Sales are also statistically significant but are negatively correlated with debt and external equity ratios and also with philanthropic ratio. Positive margin is statistically and negatively correlated to external equity but is positively correlated with philanthropic ratio.

5 Discussion and limitations

5.1 Contribution and implications

The emergence of accelerator programmes as a prominent feature in the entrepreneurial ecosystem has encouraged founders, academics, and policy makers alike to evaluate the associated benefits and costs. In this study, we focus on role of accelerator programmes in facilitating access to funding from different sources. Previous studies suggest that being part of accelerator programmes is beneficial for most ventures, as it shortens the time for raising capital and increases the amount of funding from external equity (Hallen et al., 2020; Smith & Hannigan, 2015; Smith et al., 2013). Nevertheless, these past studies fail to distinguish between the different sources of funding and neither do they evaluate the effect of business cycle conditions in moderating the relation between the capital structure itself and being accelerated. In this study, we investigate the effects of accelerator programmes on start-up's capital structure and consider the business cycle conditions over a large sample of accelerated and non-accelerated start-ups. For this purpose, we identify three financing sources, namely, debt, external equity, and philanthropic capital, which enable us to achieve a better insight into the financing behaviours of these capital providers.

In line with previous studies, we corroborate the finding that accelerator programmes have beneficial effects for start-ups (Brown et al., 2019; Hallen et al., 2020). In addition to their learning benefits (Hallen et al., 2020) and obtaining access to resources (Hallen et al., 2020; Stayton & Mangematin, 2019), accelerator programmes also facilitate access to funding from external equity investors. Compared to non-accelerated start-ups, we find that accelerated start-ups experience an increase in the percentage of funding from external equity investors, which is likely to have a positive effect on start-up performance (Wadhwa and Kotha, 2006). Additionally, our empirical analyses reveal that when the state of economy deteriorates, being accelerated can have meaningful effects for the external equity-to-capital ratio.

To explain these results, we argue that accelerator programmes mitigate agency costs and the risk for external equity investors, particularly when the economy deteriorates. In recession periods, there is a lack of capital, the asymmetry of information costs are relatively higher, and consequently the benefits of being accelerated increase. During these recessive periods, external equity investors are able to take advantage of the reduction in the value of start-ups and can accordingly invest in those start-ups which are perceived to be of better quality. Accelerated start-ups are perceived to possess better quality, as they were previously screened and filtered. During economic downturns, investors are more cautious with regard to the future of start-ups (Paik & Woo, 2013) and tend to rely more heavily on signalling effects (Epure and Guasch, 2020). Furthermore, the accelerator setup is well aligned with founders' incentives and start-ups' characteristics. In our data, accelerated start-ups are more likely to hold patents, to have an invention-based model, to have a negative profit margin, and to be founded by young and foreign entrepreneurs. These are precisely the types of founders who are more likely to benefit from a boost in credibility and the type of start-ups that are riskier and are characterised by a greater level of information asymmetry. Consequently, accelerator programmes can function as an intermediary to mitigate this risk. Nevertheless, should the number of accelerator programmes continues to growth, then different types of accelerators could well emerge, some of which could encounter difficulties in providing strong signals due to their lack of prominence, status, and credibility in the entrepreneurial landscape. It is possible that two types of accelerator programmes will emerge: (1) top tier accelerator programmes which attract the best ventures and provide a strong signal and (2) a second tier which attract weaker ventures and provide a more limited

signal of quality. This heterogeneity is likely to reduce the screening technologies of accelerator programmes and could have an adverse effect on the amount of external equity raised by accelerated start-ups.

Although accelerator programmes appear to provide information regarding the quality of the business idea and to reduce the agency costs for external equity investors, they do not, however, seem to have the same effect for debtholders and philanthropic investors. This is possibly due to the fact that debt may not be a particularly appropriate source of capital for technologically based firms, such is the case of our sample. Loans are usually subject to collateral and technology firms are based more on their intellectual value rather than physical capital (Colombo & Grilli, 2007). There is also a higher risk of failure for technologically based start-ups, which serves as an added deterrent for bank lending (Giudici & Paleari, 2000). As a result, the fact that a start-up is accelerated might not have a meaningful effect on the decision of debtholders to lend money and those entrepreneurs prefer to rely on external equity, as such investors are better prepared to absorb the cash-flow risk. Alternatively, the accelerated start-ups in our sample could well be at a stage when they are no longer young enough to participate in business plan contests and crowdfunding campaigns (philanthropic capital), and they have not yet built sufficient track record to attract debt. This argument is consistent with the life cycle theory, which contends that firms use different types of financing throughout the life of the business (Berger & Udell, 1998). Philanthropic capital is a more prominent source of funding during the first rounds of financing.

Although our empirical evidence is consistent with the mitigating risk hypothesis, we cannot discard other mechanisms, such as access to enhanced connections and networks. Accelerator programmes act as important intermediaries which facilitate contacts with mentors, industry experts, and investors alike during the acceleration process. Their unique network positively contributes to the subsequent funding of start-ups (Christiansen, 2009; Miller & Bound, 2011). Furthermore, accelerator programmes usually organise graduation events where founders pitch their start-ups to a group of potential investors, which sometimes lead to subsequent funding.

Similar to previous studies on technologically based start-ups, we find that the start-ups in our sample rely mostly on internal capital, then on external and philanthropic capital, and employ debt as a last resort (Audretsch & Lehmann, 2004; Hogan and Hutson, 2005). External and philanthropic investors appear to be attracted by the growth potential of technologically based ventures. In terms of start-ups characteristics, we find that the size of the start-up is positively correlated with the three sources of capital used in our analysis. Information asymmetries are typically less severe in larger rather than smaller start-ups, owing to the increased availability of information and the lower likelihood of default in the case of larger firms. Consistent with previous studies, we find that profit margin is negatively correlated with external equity ratio, which can be explained by the preference for internal rather than external funding, which is in line with the pecking-order theory (Brealey et al., 1977). For philanthropic investors, we find a positive correlation with the profit margin, which suggests that these investors are more likely to provide funds when the business model is profitable and scalable.

This study has implications for policy makers and entrepreneurs alike. For entrepreneurs, our findings offer insights regarding the question of whether accelerators are beneficial and how they impact the capital structure of their start-ups. Our results provide evidence that, on average, accelerators assist the procurement of venture financing, which suggests that applying for accelerated programmes could be useful for those entrepreneurs who are planning to raise equity from external investors as a first step to enhance their chance of obtaining external equity, especially during economic downturns. For policy makers, our findings suggest that the rapid expansion of the accelerator form of start-ups is useful as it helps overcome the difficulty of raising capital for new ventures.

5.2 Limitations and further research

Our study has certain limitations which could be overcome in future avenues of research. First, the origin of the data of our study is based in a self-reported survey. Although the survey administrators put in place the necessary checks for assuring data quality, such as by adopting a collection method which is consistent across the years, our study cannot rule out the possibility of bias in the self-reporting. Second, our inferences are also limited by survivor bias, where the survivor bias in our analysis originates from ventures which survived long enough to successfully attain the accelerator application stage, and which additionally survived the first 5 years of the accelerator programme. It is therefore plausible that the start-ups comprising our sample could have a higher chance of success. Third, our research design only enables us to

evaluate a start-up during three distinct periods: before the acceleration, during the acceleration, and after the acceleration. While this approach enables us to separate between accelerated and non-accelerated programmes, it is limited in terms of capturing long-term effects. Another important limitation concerns the measures available in the data. Future research would be advised to use different measures and a different design in order to better attenuate these previous limitations.

The previous studies find that not all accelerator programmes causally benefit from participating ventures (Chan et al., 2020; Cohen et al., 2019; Hallen et al., 2020). This is because accelerator programmes come in different forms and types, and it is likely that their design and setup can affect the performance of start-ups and their capital structure differently. Unfortunately, the identification of the accelerator programmes in our data is not identified in order to preserve anonymity and confidentiality. Although we account for the fixed effects of accelerator programmes in our empirical analysis, it would be interesting to rank accelerator programmes and/or to identify which characteristics are more associated with specific sources of funding. The fact that accelerator programmes can lead to different outcomes suggests an opportunity for researchers to explore the potential effects of various programme configurations on the capital structure of start-ups. Related to this issue, it is suggested that future studies should also compare the capital structure decisions of those start-ups that participated in top and second tier accelerator programmes as a way of providing further insights into the mitigating risk hypothesis.

In addition, we cannot fully disentangle mechanisms such as network access and mitigating risk. Isolating the network effects requires using different sets of data and this is certainly a topic for future research.

Finally, empirical research on accelerator programmes is limited by the two-sided sorting effects between an accelerator programme and a start-up, where the accelerator and venture preferences explain how and why an accelerator and a venture pair up and interact together. Despite our efforts to control for self-selection bias by using matching approaches, we acknowledge the limitation of our approaches.

6 Conclusion

The available empirical evidence on the financing decisions of large firms cannot be generalised to start-ups, particularly to technologically based start-ups. These start-ups differ markedly from large firms with regard to their relationship with banks and their access to credit. On the other hand, the early-stage financing of entrepreneurial ventures has experienced considerable changes in recent years. One of those changes included the widespread growth of accelerator programmes.

In this paper, we examine the capital structure decisions of accelerated and non-accelerated start-ups between 2016 and 2019 using a comprehensive worldwide accelerator database provided by the Entrepreneurship Database Program of Emory University.

We find that accelerated start-ups have a significantly higher external equity-to-capital ratios than non-accelerated start-ups, after accounting for firm-specific differences and unobserved start-ups factors. This effect remains after we control for size, profitability, age, and firm fixed effects, and it is influenced by the economic conditions of the location of the start-up. Accelerated start-ups rely more on external equity (and less on own equity) when the financial conditions deteriorate in relation to the behaviour of non-accelerated start-ups. We also confirm that accelerated start-ups raise more capital through philanthropic investors, but this effect is dissipated when we control for firm fixed effects.

Our results confirm the benefits of accelerator programmes in increasing the occurrence of new venture development and the importance of external equity investors for accelerated start-ups, particularly during recession periods. Acknowledging the fact that business acceleration has become a popular policy tool, our results support the importance of accelerator programmes for external equity investors when monitoring and screening the quality of start-ups.

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Declarations

Conflict of interest The authors declare no competing interests.

Appendix

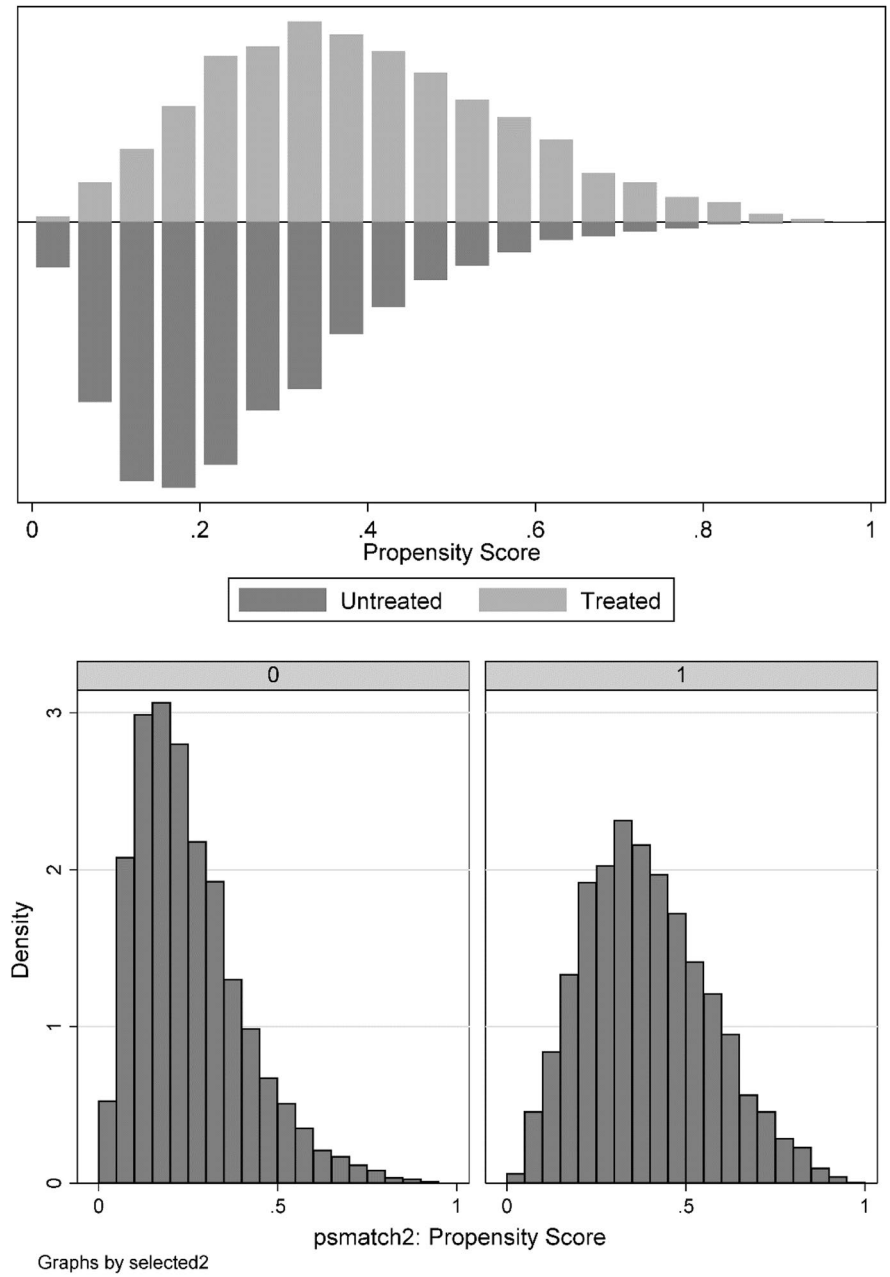


Fig. 2 Propensity score distributions between accelerated (treated) and non-accelerated (untreated) start-ups

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